THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

Royal School of Information Technology (RSIT)

Course Structure & Syllabus

(Based on National Education Policy 2020)

For

Master of Computer Applications

W.E.F

AY: 2026-2027

Course Structure and Syllabus of the Framework

6.1 Course Structure of MCA for the AY 2026-2027

Semester-I					
S.N	**Subject Code**	**Names of subjects**	**LEVEL**	**C**	**L-T-P**
1	CAP054C101	Mathematical Foundation for Computer Science	400	4	4-0-0
2	CAP054C102	Advanced Data Structures and Algorithms	499	3	3-0-0
3	CAP054C103	Advanced Database Management Systems	400	3	3-0-0
4	CAP054C104	Programming with Python	499	1	0-0-2
5	CAP054C112	Advanced Data Structures and Algorithms Lab	400	1	0-0-2
6	CAP054C113	Advanced Database Management Systems Lab	500	3	3-0-0
7	CAP054C114	Programming with Python Lab		1	0-0-2
8	CAP054C105/ CAP054C106	Introduction to Artificial Intelligence/ Introduction to Data Science	500	4	4-0-0
9	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	3 or 4		
TOTAL				**20**	
Semester-II					
S.N	**Subject Code**	**Names of subjects**	**LEVEL**	**C**	**L-T-P**
1	CAP054C201	Web Technology	499	3	3-0-0
2	CAP054C202	Modern Operating Systems	499	3	3-0-0
3	CAP054C203	Advanced Computer Networks	500	3	3-0-0
4	CAP054C211	Web Technology Lab	499	1	0-0-2
5	CAP054C212	Modern Operating Systems Lab	499	1	0-0-2
6	CAP054C213	Advanced Computer Networks Lab	500	1	0-0-2
7	CAP054C204/ CAP054C205	Introduction to Machine Learning/ Agentic AI for Data Science	500	4	4-0-0
8	CAP054C206/ CAP054C207	Data Visualisation and Interpretation/ Big Data Analytics	500	4	4-0-0
9	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	3 or 4		
TOTAL				**20**	
2nd Year Course Structure (Course Work + Research)					
Semester-III					
S.N	**Subject Code**	**Names of subjects**	**Level**	**C**	**L-T-P**
1	CAP054C301	Network Security and Cryptography	600	3	3-0-0
2	CAP054C302	Remote Sensing and GIS	600	3	3-0-0
3	CAP054C311	Network Security and Cryptography Lab	600	1	0-0-2
4	CAP054C312	Remote Sensing and GIS Lab	600	1	0-0-2
5	CAP054C303/ CAP054C304	Introduction to Deep Learning/ Social Network Analysis	600	4	4-0-0

6	CAP054C321	Dissertation-I	600	8	-
TOTAL				**20**	
Semester-IV					
S.N	**Subject Code**	**Names of subjects**	**Level**	**C**	**L-T-P**
1	CAP054C401/ CAP054C402	Generative AI and LLMs/ Blockchain Technologies	600	4	4-0-0
2	CAP054C403/ CAP054C404	Natural Language Processing/ Bioinformatics	600	4	4-0-0
3	CAP054C421	Dissertation-II	600	12	-
TOTAL				**20**	

2nd Year Course Structure (Only Research Work)

Semester-III					
S.N	**Subject Code**	**Names of subjects**	**Level**	**C**	**L-T-P**
1.	CAP054C321	Dissertation-I	600	20	-
TOTAL				**20**	

Semester-IV					
S.N	**Subject Code**	**Names of subjects**	**Level**	**C**	**L-T-P**
1.	CAP054C421	Dissertation-II	600	20	-
TOTAL				**20**	

6.1 Detailed Syllabus of 1st Semester

Paper I/Subject Name: Mathematical Foundations of Computer Science	Subject Code: CAP054C101
L-T-P-C – 3-1-0-4	Credit Units: 03	Scheme of Evaluation: T

Objective:

The objectives of the course are to enable students to develop a strong mathematical foundation for computing and problem-solving for concepts like logic, set theory, relations, functions, graph theory, combinatorics, number theory, probability, and statistical techniques.

Prerequisites: Discrete Mathematics, Linear Algebra, Probability and Statistics, Basic Understanding of Algorithms.

Course Outcomes

On successful completion of the course, the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** number theory in cryptographic algorithms.	**BT 2**
CO 2	**Apply** set theory, logic, and proof techniques in computing problems	**BT 3**
CO 3	**Analyze** and **evaluate** probabilistic models in computing, machine learning, and cryptography.	**BT 4 & 5**
CO 4	**Solve** problems related to formal languages, automata, and computational complexity.	**BT 6**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Logic, Set Theory**	Propositional Logic and Predicate Logic, Logical Connectives, Truth Tables, Normal Forms, Logical Inference, Resolution, Proof Techniques, Set Theory and Relations, Sets, Operations, Power Sets, Types of Relations: Reflexive, Symmetric, Transitive, Equivalence Relations, Functions: Injective, Surjective, Bijective, Mathematical Induction and Recursion, Inductive Proofs, Recursive Definitions and Structural Induction	**22**
II.	**Combinatorics, Graph Theory & Number Theory**	Counting Principles, Permutations and Combinations, Pigeonhole Principle, Inclusion-Exclusion Principle, Graph Theory and Applications, Graph Representation: Adjacency Matrix, Adjacency List, Eulerian and Hamiltonian Graphs, Shortest Path Algorithms (Dijkstra, Floyd-Warshall), Planar Graphs and Graph Coloring, Number Theory and Applications, Divisibility, Prime Numbers, Congruences, Fermat's Theorem, Euler's Theorem, Modular Arithmetic and Cryptography	**22**
III.	**Probability, Statistics and Randomized Algorithms**	Probability Theory, Axioms of Probability, Conditional Probability and Bayes' Theorem, Random Variables and Expectation, Statistical Methods, Mean, Variance, Standard Deviation, Probability Distributions (Binomial, Poisson, Normal), Hypothesis Testing and Confidence Intervals, Randomized	**22**

		Algorithms and Markov Chains, Monte Carlo and Las Vegas Algorithms, Markov Chains and Applications	
IV	**Formal Languages, Automata, and Computational Complexity**	Formal Languages and Automata, Regular Expressions and Finite Automata, Pushdown Automata and Context-Free Grammars, Turing Machines and Computability, Turing Machine Models, Decidability and Undecidability, Computational Complexity, P, NP, NP-Complete, and NP-Hard Problems, Approximation Algorithms and Hardness of Approximation	**22**
		TOTAL	**88**

Mathematical Foundations of Computer Science Practice Sessions

Minimum 10 Laboratory experiments based on the following-

- Implement propositional logic in Prolog.
- Write Python programs for set operations and relation properties.
- Implement graph traversal algorithms (BFS, DFS).
- Implement modular exponentiation for cryptography.
- Simulate randomized algorithms in Python.
- Implement Bayes' Theorem for spam filtering.
- Implement finite automata for pattern matching.
- Simulate Turing Machines using Python.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4*22 NCH = 88 NCH	1*30 NCH = 30 NCH	4*8 = 32 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. Discrete Mathematics and Its Applications, Kenneth H. Rosen, 7th Edition, 2017, McGraw-Hill
2. *Introduction to Algorithms,* Cormen, Leiserson, Rivest, & Stein (CLRS), 3rd Edition, 2009, MIT Press
3. *Introduction to Automata Theory, Languages, and Computation,* John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, 3rd Edition, 2008, Pearson

Reference Books:

1. Sheldon Ross, *Introduction to Probability Models*, 11th Edition, 2014, Academic Press
2. Richard Johnsonbaugh, *Discrete Mathematics*, 7th Edition, 2014, Pearson
3. Michael Sipser, *Introduction to the Theory of Computation*, 3rd Edition, 2014, Cengage
4. Narsingh Deo, *Graph Theory with Applications to Engineering and Computer Science,* New Edition, 1979, PHI

Paper II/Subject Name: Advanced Data Structure and Algorithms		Subject Code: CAP054C102
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

This course aims to provide in-depth knowledge of complex data structures and advanced algorithms, focusing on optimization techniques, real-world applications, and competitive programming skills.

Prerequisites: Basic Data Structures (Arrays, Linked Lists, Stacks, Queues), Knowledge of Sorting & Searching Algorithms, Basics of Graph Theory, and Recursion

Course Outcomes:

On successful completion of the course, the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **demonstrate** how Data Structures work.	**BT 1 & 2**
CO 2	**Apply** the Data Structures concepts to solve various problems.	**BT 3**
CO 3	**Analyze** and debug the errors while writing the programs.	**BT 4**
CO 4	**Assess** and design a new algorithm to solve a new real-life problem	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Advanced Data Structures**	Persistent Data Structures, Skip Lists, Self-balancing Trees (AVL, Red-Black, B-Trees, Splay Trees), Segment Trees, Fenwick Trees, Fibonacci Heaps, Graph Representations (Adjacency List, Adjacency Matrix, Incidence Matrix)	**22**
II	**Graph Algorithms**	Shortest Path Algorithms (Dijkstra, Bellman-Ford, Floyd-Warshall, Johnson's Algorithm), Minimum Spanning Tree (Kruskal, Prim's), Maximum Flow (Ford-Fulkerson, Edmonds-Karp), Eulerian and Hamiltonian Paths, Topological Sorting (Kahn's Algorithm, DFS-based approach)	**22**
III	**Advanced Algorithmic Techniques**	Divide & Conquer (Merge Sort, Quick Sort, Strassen's Matrix Multiplication, Closest Pair of Points), Greedy Algorithms (Huffman Coding, Activity Selection, Job Scheduling, Fractional Knapsack), Dynamic Programming (0/1 Knapsack, LCS, Floyd-Warshall, Matrix Chain Multiplication, Bellman-Ford), String Matching Algorithms - KMP, Rabin-Karp, Aho-Corasick)	**22**
IV	**Complexity & NP-Hard Problems**	Complexity Classes (P, NP, NP-Hard, NP-Complete), Reduction Techniques, Approximation Algorithms (Vertex Cover, Traveling Salesman Problem, Set Cover), Real-World Applications (AI, Bioinformatics, Game Theory, Computational Geometry,	**22**
Total			**88**

Paper III/ Subject Name: Advanced Data Structure and Algorithms Lab		Subject Code: CAP054C112
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Experiment No.	Title	Objective
1	Implementation of Linked Lists (Singly, Doubly, Circular)	Understanding dynamic memory allocation and pointer manipulation.
2	Stack & Queue Implementation	Using arrays and linked lists to implement stack and queue operations.
3	Priority Queue & Heap Implementation	Understanding heap properties and implementing Min-Heap & Max-Heap.
4	Binary Search Tree (BST) Operations	Implementing insert, delete, and search operations in BST.
5	AVL Tree Implementation	Implementing AVL rotations (Left, Right, Left-Right, Right-Left) for self-balancing.
6	Graph Representations & Traversals	Implemented adjacency list/matrix and BFS, DFS traversals.
7	Dijkstra's Algorithm for Shortest Path	Implementing Dijkstra's algorithm for weighted graphs.
8	Floyd-Warshall Algorithm	Understanding and implementing all-pairs shortest paths.
9	Kruskal's & Prim's MST Algorithms	Implementing Minimum Spanning Tree algorithms.
10	Bellman-Ford Algorithm	Understanding negative-weight edge handling in shortest path problems.
11	Topological Sorting	Implementing Kahn's algorithm and DFS-based topological sort.
12	0/1 Knapsack Problem (Dynamic Programming)	Implementing a dynamic programming approach for knapsack optimization.
13	Longest Common Subsequence (LCS) Algorithm	Implementing dynamic programming-based LCS calculation.
14	String Matching Algorithms (KMP, Rabin-Karp)	Efficient pattern searching in text processing applications.
15	Hashing Techniques & Collision Resolution	Implementing various hashing methods (Chaining, Open Addressing).
16	Segment Trees	Implementing segment trees for range queries and modifications.
17	Fenwick Trees (Binary Indexed Trees)	Understanding how to perform efficient cumulative frequency calculations.
18	Graph Coloring Problem (Backtracking)	Implementing graph coloring to solve scheduling and register allocation problems.

19	Approximation Algorithms (Vertex Cover, TSP)	Implementing heuristic-based approximation for NP-hard problems.
20	Competitive Programming Challenge	Solving real-world problems using efficient data structures and algorithms.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks:

1. Introduction to Algorithms, Cormen, Leiserson, Rivest & Stein (CLRS), 3rd Edition
2. Introduction to Data Structure, Reema Thereja, Pearson 2020

Reference Books:

1. Algorithm Design, Jon Kleinberg & Eva Tardos
2. The Art of Computer Programming, Donald Knuth
3. Competitive Programming Handbook, Antti Laaksonen

Paper III/Subject Name: Advanced Database Management Systems		**Subject Code: CAP054C103**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students learn about databases and the process of designing and constructing data models.

Prerequisites: C/C++, Concepts of Data Structures.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** various terms used in Database Management system	**BT 1**
CO 2	**Understand** the basic concepts and applications of database systems	**BT 2**
CO3	**Apply** the basic concepts to design Database	**BT 3**
CO 4	**Evaluate** the process Database design, transaction processing and concurrency control	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction**	Introduction to Data System, Drawbacks of Conventional File System, Purpose of database systems, DBMS Components, Architecture, Data Independence, Data modelling, Entity Relationship Model, Relational, Network, Hierarchical and object-oriented models, Data Modelling using the Entity Relationship Model.	**15**
II	**Relational Databases**	Relational databases, relational algebra, relational calculus. Data definition with SQL, insert, delete and update statements in SQL, views, data manipulation with SQL, triggers and assertions, cursors, Embedded SQL	**15**
III	**Normalization**	Relational Database Design guidelines, Integrity Constraints, Domain Constraints, Referential integrity, Functional Dependency, Normalization using Functional Dependencies, Normal forms (1NF, 2NF, 3NF, BCNF), Multi-valued Dependencies and Forth Normal Form, Join Dependencies and Fifth Normal Form, Pitfalls in Relational Database Design, Lossless Non-additive Join Property of Decomposition, Dependency Preserving Decomposition	**15**
IV	**Transaction Processing, Concurrency and Recovery**	Introduction. ACID Properties, Schedules and Recoverability -Serializability of Schedules- Concurrency Control, Database Recovery Concepts- Caching, Checkpoints, Transaction Rollback, Case Study-Recovery Techniques in DBMS	**15**
Total			**60**

Text Book:

1. *Fundamentals of Database System,* Elmasri, Navathe, 7th Edition, 2016, Pearson Education Asia
2. *Database System Concepts*, Korth H.F., Silberschatz A.; 6th Edition, 2013, Mc Graw Hill.
3. *Introduction to Database Management System*, Kahate A., 1st Edition, 2004, Pearson Educations
4. *DataBase Management System*, Paneerselvam, 2nd Edition, 2011, PHI Learning

Reference Books:

1. Date C.J.*, An Introduction to Database Systems,* 8th Edition, 2003, Pearson Education Asia
2. Desai B.C., *An Introduction to Database Systems,* Revised Edition, 2012, Galgotia Publications

Advanced Database Management Systems Lab

Objective:

The objectives of the course to teach the student database design and query processing through MySql.

Prerequisites: C/C++, Concepts of Data Structures

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
- Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET,
- Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.

- Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
- Creation of simple PL/SQL program which includes declaration section, executable section and exception – Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)
- Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
- Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
- Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE- APPLICATION ERROR.
- Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
- Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
- Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
- Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 22 NCH = 44 NCH	2 * 15 NCH = 30 NCH	8 * 2 NCH = 16 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Fundamentals of Database System,* Elmasri, Navathe, 7th Edition, 2016, Pearson Education Asia
2. *Database System Concepts*, Korth H.F., Silberschatz A., 6th Edition, 2013, Mc Graw Hill.

Reference Books:

1. Date C.J.*, An Introduction to Database Systems,* 8th Edition, 2003, Pearson Education Asia
2. Desai B.C., *An Introduction to Database Systems,* Revised Edition, 2012, Galgotia Publications

Paper IV/Subject Name: Programming with Python		**Subject Code: CAP054C104**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to teach the students techniques to build and develop Python codes.

Prerequisites: Fundamentals of Computer Programming

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**

CO 1	**Understand** the basic concepts and terminologies of Python Programming	**BT 2**
CO 2	**Apply** the concepts learnt to write efficient programs.	**BT 3**
CO 3	**Analyze** and **evaluate** the codes to fix the errors	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Algorithmic Problem Solving**	Algorithms, building blocks of algorithms (statements, state, control flow, functions), notation (pseudo code, flow chart, programming language), algorithmic problem solving, simpler strategies for developing algorithms (iteration, recursion).	**15**
II	**Data, Expressions, Statements**	Python interpreter and interactive mode; values and types: int, float, boolean, string, and list; variables, expressions, statements, tuple assignment, precedence of operators, comments; modules and functions, function definition and use, flow of execution, parameters and arguments	**18**
III	**Control Flow, Functions**	Conditionals: Boolean values and operators, conditional (if), alternative (if-else), chained conditional (if-elif-else); Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays. .	**18**
IV	**Lists, Tuples, Dictionaries**	Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods; advanced list processing – list comprehension	**15**
Total			**66**

Programming with Python Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Write a program to add two numbers.
2. Write a program that declares 3 integers, determines and prints the largest and smallest in the group.
3. Write a program for the factorial of a number.
4. Write a program to calculate simple interest.
5. Write a program to find out whether a given year is a leap year or not.
6. Write a program to implement linear search and binary search.
7. Write a program to find that given number is Armstrong or not.
8. Write a program to print the Fibonacci Series.
9. Write a program to convert a decimal number into binary numbers.
10. Python Program to find the sum of an array.
11. Write a program to find the largest number of elements in the array.

12. Write a program to check if a string is a palindrome or not.
13. Maintain book records as per their serial numbers in the library using a dictionary.
14. Write a program to concatenate two dictionaries into one.
15. Perform the following operations on the dictionary: 1) Insert, 2) delete, 3) change, and 4) update.
16. Write a program to calculate the addition of two numbers using methods.
17. Program to calculate the average of numbers using function.
18. Fibonacci series using recursion.
19. Write a program to create a module of factorial in Python.
20. Write A Program to Find the Area of a Rectangle Using Classes
21. Write A Program to Append, Delete, and Display Elements of a List Using Classes
22. Write A Program to Create a Class and Compute the Area and the Perimeter of the Circle
23. Write A Program to Create a Class that Performs Basic Calculator Operations
24. Write A Program to Create a Class in which One Method Accepts a String from the User and
25. Another Prints it.
26. Write A Program that Reads a Text File and Counts the Number of Times a Certain Letter
27. Appears in the Text File.
28. Write A Program to Read a Text File and Print all the Numbers Present in the Text File.
29. Write programs to visualize data using NumPy
30. Write programs to analyse data using NumPy

National Credit Hours		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 2 NCH = 16 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *An Introduction to Python – Revised and updated for Python 3.2,* Guido van Rossum and Fred L. Drake Jr, 2011, Network Theory Ltd

Reference Books:

1. Timothy A. Budd, *Exploring Python*, 2015, Mc-Graw Hill Education (India) Private Ltd.
2. Kenneth A. Lambert, *Fundamentals of Python: First Programs*, 2nd Edition, 2012, CENGAGE Learning.
3. Charles Dierbach, *Introduction to Computer Science using Python: A Computational Problem- Solving Focus*, 1st Edition, 2013, Wiley India Edition
4. Paul Gries, Jennifer Campbell, and Jason Montojo, *Practical Programming: An Introduction to Computer Science using Python 3*, 2nd edition, 2013, Pragmatic Programmers, LLC.

Paper V/ Subject Name: Introduction to Artificial Intelligence		**Subject Code: CAP054C105**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand fundamental AI concepts and core AI techniques, explore machine learning and neural networks as key AI components, etc.

Prerequisites: Fundamentals of Propositional Logic, mathematics.

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain the** fundamental concepts, applications, and ethical implications of AI.	**BT 2**
CO 2	**Apply** uninformed and informed search algorithms to solve AI problems	**BT 3**
CO 3	**Analyze** and implement knowledge representation techniques, including logic-based and probabilistic reasoning.	**BT 4**
CO 4	**Assess** and **design** AI-based solutions using reasoning, decision-making, and planning techniques.	**BT 5 & 6**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I.	**Introduction**	Definition, History, and Evolution of AI, Applications of AI (Healthcare, Finance, Robotics, NLP, etc.), AI vs. Machine Learning vs. Deep Learning, Strong AI vs. Weak AI, AI as Search: Problem Formulation, State-Space Representation, Rational Agents, Types of Agents, Breadth-First Search (BFS), Depth-First Search (DFS), Depth-Limited Search & Iterative Deepening DFS, Uniform Cost Search, Heuristic Function & Admissibility, Greedy Best-First Search, A* Algorithm (Manhattan, Euclidean Heuristics), Hill Climbing & Local Search Algorithms, Definition and Examples (Sudoku, N-Queens), Backtracking Algorithm, Constraint Propagation: Forward Checking, Arc Consistency (AC-3)	**22**
II.	**Knowledge Representation & Reasoning**	Types of Knowledge: Declarative vs. Procedural, Common Sense Knowledge, Knowledge-Based Systems, Propositional Logic: Syntax, Semantics, Logical Connectives, Truth Tables, First-Order Logic (FOL): Predicates, Functions, Quantifiers, Unification & Resolution in FOL, Forward Chaining vs. Backward Chaining, Expert Systems & Case Study: MYCIN (Medical Diagnosis System), Bayesian Networks: Structure, Conditional Probability Tables (CPT), Exact & Approximate Inference in Bayesian Networks, Hidden Markov Models (HMM), Fuzzy Sets, Membership Functions Fuzzy Inference Systems (Mamdani & Sugeno), Defuzzification Techniques	**22**
III.	**Planning in AI**	Definition of Planning in AI, STRIPS Representation and PDDL (Planning Domain Definition Language), State-Space Search in Planning, Forward & Backward Planning, Partial Order Planning (POP), Graph Plan Algorithm, Decision Trees & Utility Theory, Game Theory in AI, Adversarial Search: Minimax Algorithm & Alpha-Beta Pruning, MDP Formulation, Bellman Equations, Policy Evaluation & Policy Iteration, Q-Learning Algorithm	**22**
IV	**AI Applications**	NLP: Text Processing & Tokenization, Named Entity Recognition (NER), Sentiment Analysis Computer Vision: Image Classification & Object Detection, Feature Extraction Techniques Reinforcement Learning: Deep Q-Learning & Neural Networks in RL, Case Study: AI for Self-Driving Cars	**22**

		AI Bias & Fairness, Explainable AI (XAI), AI for Social Good	
TOTAL			**88**

Artificial Intelligence Practice Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Implement BFS & DFS in Python
- Solve a pathfinding problem using A* Search
- Constraint satisfaction solver for Sudoku
- Implement logical inference using Propositional Logic.
- Build a Rule-Based Expert System for disease diagnosis.
- Implement a Bayesian Network for predicting weather conditions.
- Develop a Fuzzy Logic Controller for temperature regulation.
- Implement STRIPS-based AI Planning for a block-stacking problem.
- Develop a Tic-Tac-Toe AI using Minimax Algorithm.
- Implement Q-Learning for a simple game (Grid World).
- Sentiment Analysis on Twitter Data using NLP.
- Implement a Handwritten Digit Classifier using OpenCV.
- Train an AI model using Q-Learning for a custom environment.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. *Artificial Intelligence: A Modern Approach,* Stuart Russell & Peter Norvig, 4th Edition, 2020, PHI
2. *Artificial Intelligence,* Elaine Rich, Kevin Knight, Shivashankar B. Nair, 3rd Edition, 2017, Tata McGraw Hill

Reference Books:

1. Nils J. Nilsson, *Principles of Artificial Intelligence,* 1993, Morgan Kaufmann Publishers

Paper V/Subject Name: Introduction to Data Science		**Subject Code: Cap054C106**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:
The objective of this course is to introduce students to data science fundamentals, including data collection, preprocessing, exploratory analysis, statistical modeling, and basic machine learning techniques. The focus will be on structured and semi-structured data before transitioning to big data analytics in the next course.

Prerequisites: Basic computer programming, fundamental mathematics

Course Outcomes

On successful completion of the course the students will be able to:

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Understand the fundamental concepts and scope of Data Science	**BT 1 & 2**
CO 2	Perform data collection, cleaning, and preprocessing for structured and semi-structured data	**BT 3**
CO 3	Conduct exploratory data analysis (EDA) and apply statistical techniques for insight generation	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction to Data Science & Data Types**	Overview of Data Science, Key applications in real-world industries, Structured vs Semi-Structured vs Unstructured data, Data Science Workflow, Tools & Technologies (Python/R, Jupyter, SQL, Excel, BI Tools)	**12**
II	**Data Collection, Preprocessing & Wrangling**	Data collection techniques (APIs, Web Scraping, Databases), Handling missing values, Outlier detection, Encoding categorical variables, Data Transformation (Scaling, Normalization, Feature Engineering)	**12**
III	**Exploratory Data Analysis (EDA) & Statistical Techniques**	Descriptive Statistics, Data Visualization (Histograms, Box plots, Pair plots, Heatmaps), Correlation & Covariance, Hypothesis Testing (t-test, ANOVA, Chi-Square test), Feature Selection	**12**
IV	**Machine Learning for Structured Data**	Introduction to Supervised & Unsupervised Learning, Linear & Logistic Regression, Decision Trees, Model Evaluation (Accuracy, Precision, Recall, F1-Score, RMSE), Preparing for Big Data Analytics (Data Pipeline Concepts)	**12**
TOTAL			**48**

National Credit Hours		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 80 NCH	-	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Introduction to Data Science Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Installation and setup of Anaconda, Jupyter Notebook, and Google Colab.

- Introduction to NumPy arrays and mathematical operations.
- Data manipulation using Pandas DataFrames.
- Reading and writing data from CSV, Excel, and JSON files
- Collecting data from public datasets and APIs.
- Handling missing values using deletion and imputation methods.
- Detecting and treating outliers.
- Data transformation, normalization, and standardization.
- Encoding categorical variables and feature engineering.
- Descriptive statistical analysis using Pandas.
- Data visualization using Matplotlib and Seaborn.
- Histograms, box plots, scatter plots, and pair plots.
- Correlation analysis and heatmap generation.
- Exploratory analysis on a real-world dataset.
- Implementation of t-test and Chi-Square test.
- ANOVA for comparing multiple groups.
- Feature selection using statistical methods.
- Linear Regression for prediction problems.
- Logistic Regression for classification problems.
- Decision Tree classification and regression.
- Model evaluation using Accuracy, Precision, Recall, F1-Score, and RMSE.
- Mini-project: End-to-end data science workflow on a real-world dataset.

Textbook:

1. Doing Data Science, Cathy O'Neil & Rachel Schutt, 2013

Reference Books:

1. Data Science for Business, Foster Provost & Tom Fawcett, 2013
2. Python for Data Analysis, Wes McKinney, 2017
3. Introduction to Statistical Learning, Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, 2013
4. Data Science from Scratch, Joel Grus, 2019

6.3 Detailed Syllabus of 2nd Semester

Paper I/ Subject Name: Web Technology		Subject Code: CAP054C201
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

To provide a comprehensive understanding of the modern web ecosystem, including client- and server-side technologies. Students will learn to design responsive web interfaces, develop backend logic, integrate APIs, and deploy secure full-stack applications. The course also introduces modern development practices like CI/CD and cloud-based deployment.

Prerequisites:

- Basic programming knowledge (preferably Python or JavaScript)
- Familiarity with HTML and CSS is recommended
- Basic understanding of how the internet and web browsers work

Course Outcomes:

SI No	Course Outcome	Bloom's Taxonomy Level
CO 1	Understand core web technologies and develop interactive web pages	BT 1 & 2
CO 2	Implement web development using modern front-end and back-end frameworks	BT 3
CO 3	Develop dynamic web applications using RESTful APIs and cloud integration	BT 4
CO 4	Analyze and apply secure web development practices	BT 5
CO 5	Design full-stack web applications following industry standards	BT 6

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction to Internet and Web Page Design**	Introduction to Web Technologies: Evolution of the Web, Web 2.0 & Web 3.0, Web Standards (W3C, ECMA). HTML5 & CSS3: Semantic elements, Forms, Flexbox, Grid Layout, Media Queries, Animations, Transitions. JavaScript & ES6+: DOM Manipulation, Async/Await, Fetch API, Event Handling, JSON, Promises & Callbacks. Front-End Frameworks: React.js (Components, Props, State Management, Hooks), Tailwind CSS, Bootstrap	22
II	**Web Browsers, Markup Language Basics, and XML**	Web Browsers: functions and working principle of web browsers; plug-ins & helper applications; conceptual architecture of some typical web browsers. Markup language basics: Standard Generalized Markup Language (SGML)- structures, elements, Content models, DTD, attributes, entities. Extensible Markup Language (XML): Markup Languages: HTML5, XML, JSON, SVG. Data Handling in XML & JSON: XML Schema, XSLT, JSON Schema, Fetching & Parsing APIs. API Development: RESTful API vs GraphQL, API Authentication & Rate Limiting	22

III	**Web Server Side**	Web Servers: Architecture of Web Servers, Apache, Nginx, Node.js Express, Serverless Web Apps. Back-End Development: Node.js, Express.js, Flask/Django, Database Integration (MongoDB, MySQL, Firebase, PostgreSQL). Authentication & Authorization: JWT, OAuth2, Role-Based Access Control (RBAC). Cloud Integration: AWS Lambda, Firebase Functions, Cloud Storage Solutions (S3, Google Cloud Storage), Deployment Strategies (Docker, Kubernetes).	**22**
IV	**Advanced Web Technologies and Web Security**	Modern Web Technologies: Progressive Web Apps (PWAs), WebSockets, WebRTC, Headless CMS (Contentful, Strapi). Cloud & DevOps: Serverless Computing, Infrastructure as Code (Terraform, CloudFormation), CI/CD Pipelines (Jenkins, GitHub Actions). Web Security: HTTPS, CSP, CORS, OWASP Top 10 Vulnerabilities, Secure Coding Best Practices, API Security (Rate Limiting, Token-Based Authentication), SQL Injection & Cross-Site Scripting (XSS) Prevention	**22**
Total			**88**

Subject Name: Web Technology		**Subject Code: CAP054C211**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 15 Laboratory experiments based on the following-

Experiment No.	Title	Objective
1	Responsive Web Design with HTML5 & CSS3	Develop a responsive website using CSS Flexbox & Grid
2	JavaScript Event Handling & DOM Manipulation	Implement dynamic UI updates using JavaScript
3	RESTful API Development	Build a REST API with Node.js, Express & MongoDB
4	Frontend Development with React.js	Create a Single Page Application (SPA) using React
5	User Authentication with JWT & OAuth2	Implement secure login & authentication in a web app
6	Cloud Deployment of Web Applications	Deploy a web app on AWS Lambda or Firebase
7	API Security with Rate Limiting	Secure APIs using JWT, OAuth2 & API Gateways
8	WebSockets & Real-Time Communication	Develop a real-time chat application with WebSockets
9	Implementing CI/CD Pipelines	Automate web deployment with GitHub Actions & Docker
10	Progressive Web Application (PWA)	Build a PWA with offline support using Service Workers
11	Secure Web Development	Prevent SQL Injection, XSS, and CSRF attacks in web apps
12	Headless CMS Integration	Connect React/Next.js with a headless CMS (Concertful/Strapi)
13	Web Performance Optimization	Analyze website performance using Lighthouse & Chrome DevTools
14	Containerized Web Applications	Deploy a full-stack application using Docker & Kubernetes
15	WebAssembly (WASM) Integration	Run high-performance code using WebAssembly with JavaScript

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 22 NCH = 44 NCH	2 * 15 NCH = 30 NCH	8 * 2 NCH = 16 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbook:

1. *Web Development and Design Foundations with HTML5*, Terry Felke-Morris, 9th Edition, 2020, Pearson.

Reference Books:

1. *Full Stack Web Development with React and Node.js*, David Griffiths, 1st Edition, 2021, O'Reilly Media.
2. *Eloquent JavaScript: A Modern Introduction to Programming*, Marijn Haverbeke, 3rd Edition, 2018, No Starch Press.
3. *Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics*, Jennifer Robbins, 5th Edition, 2018, O'Reilly Media.

Paper II/ Subject Name: Modern Operating Systems		**Subject Code: CAP054C202**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to introduce the students with the concept of Operating Systems, Process Management, CPU, Deadlocks, and Memory Management, Storage and Input/ Output Management, Protection, Security, Virtual machines and Distributed Systems.

Prerequisites: Basic understanding of computer organization

Course Outcomes

On successful completion of the course, the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the architecture, organization, functions, and services of operating systems	**BT 2**
CO 2	**Apply** process management techniques, CPU scheduling algorithms, multithreading concepts, process synchronization mechanisms, and deadlock handling strategies to solve operating system problems.	**BT 3**
CO 3	**Analyze** memory management schemes, virtual memory techniques, and file system organization to evaluate resource utilization and system performance.	**BT 4**
CO 4	**Evaluate** different operating system algorithms and resource management techniques	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Operating Systems Overview**	Computer System Overview-Basic Elements, Instruction Execution, Interrupts, Memory Hierarchy, Cache Memory, Direct Memory Access, Multiprocessor and Multicore Organization. Operating system overview-objectives and functions, Evolution of Operating System.- Computer System Organization Operating System Structure and Operations- System Calls, System Programs, OS Generation and System Boot, OS Operations, Kernel Data Structures, OS Services	**22**
II	**Process Management**	Processes – Process Concept, Process Scheduling, Operations on Processes, Inter-process Communication; CPU Scheduling – Scheduling criteria, Scheduling algorithms, Multiple-processor scheduling, Real time scheduling; Threads- Overview, Multithreading models, Threading issues; Process Synchronization – The critical-section problem, Synchronization hardware, Mutex locks, Semaphores, Classic problems of synchronization, Critical regions, Monitors; Deadlock – System model, Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.	**22**
III	**Memory Management and File Systems**	Main Memory – Background, Swapping, Contiguous Memory Allocation, Paging, Segmentation, Segmentation with paging, 32 and 64 bit architecture Examples; Virtual Memory – Background, Demand Paging, Page Replacement, Allocation, Thrashing; Allocating Kernel Memory. File-System Interface- The concept of Files, Access Methods, Directory and Disk Structure, File-System Mounting, File Sharing, Protection. File-System Implementation- File-System Structure, Directory Implementation, Allocation Methods, Free-Space Management, Disk Scheduling, Disk Management, Swap-Space Management, RAID Structure, Stable-Storage.	**22**
IV	**Case Study**	OS Security, Unix Primer, Search and sort tools, AWK tool in Unix, Unix Kernel Architecture, Shell Script in Unix, AWK tool in Unix, Make tool in UNIX, System Administration in Unix, Source Code control system in Unix, X Windows in Unix, Linux System – Design Principles, Kernel Modules, Process Management, Scheduling, Memory Management, Input-Output Management, File System, Inter-process Communication;	**22**
	TOTAL		**88**

Modern Operating Systems Practical Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Laboratory experiments based on the following-

1. Introduction- OS Structure, OS Operations, Shell Scripts, Kernel Data Structures.
2. Operating-System Structures- OS Services, User Interface to OS, System Calls, System Programs.
3. Processes- Process Concept, Process Scheduling, Operations on Processes, Interprocess Communication, Communication in Client–Server Systems.
4. Threads- Overview, Multithreading Models, Thread Libraries, Implicit Threading, Threading Issues.
5. Process Synchronization- Introduction, Mutex Locks, Semaphores.
6. CPU Scheduling- Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Thread Scheduling, Multiple-Processor Scheduling, Real-Time Scheduling.
7. Deadlocks- Algorithms for Deadlock Detection/Prevention/Avoidance/Recovery.
8. File-System Interface- The concept of Files, Access Methods, Directory and Disk Structure, File-System Mounting, File Sharing, Protection.
9. The student will perform a project to reflect software project cycles

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Operating System Concepts,* Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, 7th Edition, 2006, Wiley.

Reference Books:

1. William Stallings, *Operating Systems: Internals and Design Principles*, 7th Edition, 2013, Pearson Education India.
2. Andrew S. Tanenbaum and Herbert Bos, *Modern Operating Systems*, 4th Edition, 2014, Pearson Education India
3. Achyut Godbole and Atul Kahate, *Operating Systems*, 3rd Edition, 2017, McGraw Hill Education.

Paper III/ Subject Name: Advanced Computer Networks		**Subject Code:CAP054C203**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

To provide a comprehensive understanding of network design principles and protocols used in the Internet. The course covers layered architectures, addressing schemes, routing protocols, IP design, subnetting, switching techniques, and security considerations to equip students with practical skills for designing and implementing scalable, efficient, and secure networks.

Prerequisites: Basic knowledge of computer networks, Familiarity with TCP/IP model and networking hardware

Course Outcomes:

SI No	Course Outcome	Bloom's Taxonomy Level
CO 1	Understand Internet architecture and layered protocol models	**BT 1 & 2**
CO 2	Design and implement IPv4/IPv6 addressing and subnetting schemes	**BT 3 & 4**
CO 3	Analyze and configure routing protocols for network design	**BT 4 & 5**
CO 4	Apply switching, NAT, and VPN concepts to network design	**BT 5**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I	**Introduction to Internet Protocols**	Overview of network architecture and Internet standards. OSI and TCP/IP models. IP addressing: IPv4, IPv6, private/public addresses, address resolution (ARP, RARP), ICMP. Subnetting and supernetting.	22
II	**Routing and Switching**	Routing concepts, static and dynamic routing, distance vector and link state protocols (RIP, OSPF, BGP). Switching: LAN switching, VLANs, inter-VLAN routing, spanning tree protocol (STP), link aggregation.	22
III	**Network Design Principles**	Hierarchical network design, addressing plans, redundancy and failover, NAT, DHCP, DNS, wireless networks, enterprise topologies, ISP-level networking, remote access (VPN, tunneling).	22
IV	**Security and Performance**	Network security fundamentals: firewalls, IDS/IPS, packet filtering. Secure protocols (HTTPS, SSH, IPSec). Monitoring and troubleshooting: SNMP, NetFlow, Wireshark. Performance optimization and QoS.	22

Subject Name: Advanced Computer Networks Lab		**Subject Code:CAP054C213**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Advanced Computer Networks Lab (4 Hours/Week)

1. Design and implement IP addressing schemes for a given scenario
2. Configure static routing between multiple routers using packet tracer
3. Implement dynamic routing using RIP and OSPF
4. Setup and configure VLANs and inter-VLAN routing
5. Configure a basic firewall using access control lists (ACLs)
6. Implement DHCP and DNS in a simulated environment
7. Design and simulate a hierarchical enterprise network topology
8. Analyze packet transmission using Wireshark
9. Implement NAT and port forwarding
10. Configure secure remote access using SSH
11. Setup and test a site-to-site VPN tunnel
12. Use SNMP and NetFlow for network monitoring

13. Simulate IPv6-based network design

14. Configure wireless network access and MAC filtering

15. Analyze routing tables and troubleshoot network loops

16. Create redundancy using HSRP or VRRP

17. Optimize network performance using QoS policies

18. Secure a router with passwords and encryption

19. Monitor network using syslog and SNMP traps

20. Mini project: Design and simulate a secure, scalable network for a multi-branch organization

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	2 * 15 = 30 NCH	45 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Computer Networks,* Tannenbaum, 5th Edition, 2013, Pearson Education.
2. *Data and Computer Communication*, William Stallings, 9th Edition, 2011, Pearson Education, Inc.

Reference Books:

1. Larry L. Peterson and Bruce S. Davie, *Computer Networks: A System Approach,* 5th Edition, 2012, Morgan Kaufmann, Elsevier, 2012.
2. Behrouz A. Forouzan, *Data Communications and Networking,* 5th Edition, 2017, McGraw Hill

Paper IV/ Subject Name: Introduction to Machine Learning		**Subject Code: CAP054C204**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to introduce the fundamental concepts, workflow, and applications of Machine Learning, develop the ability to preprocess data and implement supervised and unsupervised learning algorithms, enable students to evaluate, compare, and interpret machine learning models and provide hands-on experience in developing machine learning solutions using Python and Scikit-learn.

Prerequisites: Probability & Statistics, Linear Algebra, Python

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the concepts, workflow, and applications of machine learning, including data preprocessing techniques	**BT 2**

CO 2	**Apply** supervised and unsupervised machine learning algorithms to solve real-world problems.	**BT 3**
CO 3	**Analyze** and **assess** machine learning models using appropriate performance metrics and validation techniques.	**BT 4 & 5**
CO 4	**Design** and present complete machine learning solutions using Python while considering ethical and practical deployment aspects.	**BT 6**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I.	**Introduction**	Introduction to Artificial Intelligence, Data Science and Machine Learning, Types of Machine Learning (Supervised, Unsupervised, Reinforcement), Applications of ML, Machine Learning workflow, Dataset preparation, Feature and target variables, Data preprocessing, Handling missing values, Encoding categorical variables, Feature scaling, Train-Test split, Introduction to Python libraries (NumPy, Pandas, Scikit-learn).	**22**
II.	**Supervised Learning**	Regression concepts, Linear Regression, Multiple Linear Regression, Polynomial Regression (overview), Classification concepts, Logistic Regression, K-Nearest Neighbors (KNN), Decision Trees, Random Forest, Support Vector Machine (concept), Bias-Variance Tradeoff, Model overfitting and underfitting.	**22**
III.	**Unsupervised Learning and Model Evaluation**	Clustering concepts, K-Means Clustering, Hierarchical Clustering, DBSCAN (overview), Association Rule Mining (Apriori concept), Dimensionality Reduction: PCA (concept), Model evaluation metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, Cross-validation, Hyperparameter tuning using Grid Search.	**22**
IV	**Advanced Concepts and Applications**	Introduction to Neural Networks and Deep Learning, Ensemble Learning (Bagging, Boosting, AdaBoost, Gradient Boosting), Explainable AI (Feature Importance, SHAP overview), Ethics and Fairness in Machine Learning, ML deployment overview, Real-world applications in healthcare, finance, agriculture, cybersecurity, and recommendation systems, Mini Case Studies.	**22**
TOTAL			**88**

Introduction to Machine Learning Practice Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Installation of Python environment, Jupyter Notebook, and Scikit-learn; introduction to datasets.
- Data preprocessing using Pandas: handling missing values, encoding, scaling, and train-test splitting.
- Exploratory Data Analysis (EDA) using Matplotlib and Seaborn.
- Implementation of Linear Regression on a real-world dataset.
- Implementation of Logistic Regression for classification problems.
- Classification using K-Nearest Neighbors (KNN).
- Decision Tree and Random Forest classifiers with performance comparison.

- Clustering using K-Means and visualization of clusters.
- Principal Component Analysis (PCA) for dimensionality reduction.
- Model evaluation using confusion matrix, ROC curve, precision, recall, and F1-score. 2
- Hyperparameter tuning using GridSearchCV and Cross-validation. 2
- Mini Project: Develop and evaluate a machine learning model on a real-world dataset and present the findings.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks

1. *Pattern Recognition and Machine Learning,* Christopher M. Bishop, 1st Edition, 2006, Springer
2. *Deep Learning,* Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2016, MIT Press

Reference Books:

1. Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, 2012, MIT Press
2. Richard S. Sutton, Andrew G. Barto, *Reinforcement Learning: An Introduction,* 2nd Edition, 1998, Bradford Books
3. Michael Nielsen, *Neural Networks and Deep Learning*, 2010

Paper IV/ Subject Name: Agentic AI for Data Science		**Subject Code: CAP054C205**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the concepts and architecture of Agentic AI systems in data science, apply Large Language Models (LLMs) and AI agents to automate data science tasks., develop intelligent workflows for data preparation, analysis, visualization, and predictive modeling and design responsible and trustworthy AI-powered data science applications.

Prerequisites: Probability & Statistics, Data Science

Course Outcomes

On successful completion of the course, the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the concepts, architecture and applications of Agentic AI in the data science lifecycle.	**BT 2**
CO 2	**Apply** AI agents and LLM-based techniques to automate data preprocessing, exploration and analytics tasks.	**BT 3**
CO 3	**Analyze** and **assess** agent-based machine learning workflows using appropriate performance and explainability techniques.	**BT 4 & 5**
CO 4	**Design** and implement responsible Agentic AI solutions for end-to-end data science applications.	**BT 6**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I.	**Introduction**	Evolution of AI, Generative AI and Agentic AI; Characteristics of AI agents; Components of an AI agent (Reasoning, Planning, Memory and Tool Use); Overview of Large Language Models (LLMs); Data Science lifecycle; Role of Agentic AI in data collection, preprocessing, analytics and decision support; Applications in healthcare, finance, business intelligence, agriculture and cybersecurity	**22**
II.	**AI Agents for Data Preparation and Analytics**	Intelligent data collection, Data preprocessing using AI agents, Missing value treatment, Feature engineering, AI-assisted Exploratory Data Analysis (EDA), Automated data visualization, Prompt engineering for data analytics, Retrieval-Augmented Generation (RAG) for knowledge-assisted analytics, Vector embeddings and vector databases, AI-assisted SQL generation and data querying.	**22**
III.	**Agentic Machine Learning Workflows**	AI-assisted model selection, AutoML concepts, Agent-based machine learning pipelines, Model evaluation and hyperparameter optimization, Explainable AI (SHAP and feature importance), Multi-agent collaboration for analytics, Intelligent reporting, AI copilots for data scientists, Workflow orchestration using LangChain/LangGraph/CrewAI (concepts).	**22**
IV	**Deployment, Governance and Responsible AI**	Designing autonomous analytics systems, Human-in-the-loop AI, Monitoring AI agents, AI governance, Privacy and security in Agentic AI, Hallucination detection and mitigation, Bias and fairness, Ethical AI, Deployment of AI-powered analytics applications, Future trends in autonomous data science.	**22**
TOTAL			**88**

Agentic AI for Data Science Practice Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Setup Python environment and explore LLM APIs, Pandas, Scikit-learn, LangChain/LangGraph libraries.
- Build an AI-assisted data preprocessing pipeline for handling missing values, encoding, scaling and outlier detection.
- Perform AI-assisted Exploratory Data Analysis (EDA) with automated visualization and summary generation.
- Create prompts for dataset exploration, statistical analysis and visualization.
- Develop a Retrieval-Augmented Generation (RAG) application for querying structured datasets.
- Build an AI agent that generates SQL queries and retrieves information from a database.
- Develop an AI copilot for feature engineering and model recommendation.
- Implement an AutoML workflow and compare multiple machine learning models.
- Generate explainable AI reports using SHAP or feature importance techniques.
- Build an intelligent dashboard integrating AI-generated insights using Streamlit or Dash.
- Develop a multi-agent workflow for end-to-end data analytics (ingestion, analysis, visualization and reporting).
- Mini Project: Design and implement an Agentic AI system that automates a complete data science workflow on a real-world dataset.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks

1. *AI Engineering: Building Applications with Foundation Models,* Chip Huyen, 2025, O'Reilly
2. *Introduction to Machine Learning with Python,* Andreas C. Müller and Sarah Guido, 1st Edition, 2016, O'Reilly.

Reference Books:

1. Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, 2012, MIT Press
2. Richard S. Sutton, Andrew G. Barto, *Reinforcement Learning: An Introduction,* 2nd Edition, 1998, Bradford Books
3. Michael Nielsen, *Neural Networks and Deep Learning*, 2010

Paper V/ Subject Name: Data Visualisation and Interpretation		**Subject Code: CAP054C206**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to enable students to understand the principles and techniques of effective data visualization, apply visualization methods to explore and interpret structured and unstructured data, develop interactive dashboards and communicate analytical insights using modern visualization tools.

Prerequisites: Probability & Statistics, Python Programming

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the characteristics of big data and the challenges associated with its storage and processing.	**BT 2**
CO 2	Apply Hadoop architecture, HDFS, and MapReduce programming for distributed data processing.	**BT 3**
CO 3	Apply Hadoop architecture, HDFS, and MapReduce programming for distributed data processing.	**BT 4 & 5**
CO 4	Design and optimize NoSQL database systems (e.g., MongoDB, Cassandra) for big data applications.	**BT 6**

Detailed Syllabus:

Module	**Topics**	**Course Content**	**Periods**

I.	**Introduction**	Introduction to Data Visualization, Importance of visual analytics, Types of data (categorical, numerical, temporal, spatial), Data preprocessing for visualization, Principles of visual perception, Gestalt principles, Color theory, Chart selection and design guidelines, Introduction to visualization tools: Matplotlib, Seaborn, Plotly, Tableau and Power BI.	**22**
II.	**Exploratory Data Analysis and Statistical Interpretation**	Descriptive statistics, Data distributions, Histograms, Boxplots, Scatter plots, Pair plots, Heatmaps, Correlation and covariance analysis, Outlier detection, Multivariate visualization, Dimensionality reduction (PCA overview), Interpretation of trends, patterns and anomalies, Data quality assessment.	**22**
III.	**Interactive Visualization and Business Intelligence**	Interactive visualizations using Plotly, Dash and Streamlit, Dashboard design principles, KPI visualization, Business Intelligence concepts, Tableau and Power BI dashboards, Time-series visualization, Geospatial visualization, Network visualization, AI-assisted visualization, Visual analytics for decision support.	**22**
IV	**Data Storytelling, Ethics and Applications**	Data storytelling principles, Communicating insights to technical and non-technical audiences, Infographics, Visual reporting, Explainable AI visualization (SHAP and feature importance overview), Ethical visualization, Bias and misleading graphs, Accessibility in visualization, Case studies in healthcare, finance, education, e-commerce and smart cities.	**22**
TOTAL			**88**

Data Visualisation and Interpretation Practice Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Installation and familiarization with Python visualization libraries (Matplotlib, Seaborn, Plotly).
- Data preprocessing and exploratory analysis using Pandas.
- Create histograms, bar charts, line charts, pie charts and boxplots for different datasets.
- Correlation analysis and heatmap visualization.
- Scatter plots, pair plots and multivariate visualization.
- Time-series visualization and trend analysis.
- Interactive visualization using Plotly.
- Dashboard development using Dash or Streamlit.
- Dashboard creation using Tableau Public or Microsoft Power BI.
- Visualization of machine learning model performance (Confusion Matrix, ROC Curve, Precision-Recall Curve).
- AI-assisted data visualization and automated report generation using LLM tools.
- Mini Project: Develop an interactive dashboard and present insights from a real-world dataset.

Credit Distribution

Lecture/ Tutorial	Practicum	Experiential Learning
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbook

1. *Data Warehousing in the Age of Big Data*, K.N. Krishnan, Illustrated Edition, 2013, Morgan Kaufmann Pub
2. *Python for Data Analysis*, Wes McKinney, 2013, O'Reily Media

Reference Books

1. Cole Nussbaumer Knaflic, *Storytelling with Data,* 1st Edition, 2015, Wiley
2. Gareth James et al., *An Introduction to Statistical Learning*, 1st Edition, 2017, Springer

Paper V/ Subject Name: Big Data Analytics		**Subject Code: CAP054C207**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to enable students to understand the fundamentals of big data and its challenges, learn big data processing techniques and tools, apply machine learning techniques to big data, and develop big data solutions for real-world applications.

Prerequisites: Probability & Statistics, Database Management Systems (DBMS), Python/Java Programming, Basic Data Structures and Algorithms

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the characteristics of big data and the challenges associated with its storage and processing.	**BT 2**
CO 2	Apply Hadoop architecture, HDFS, and MapReduce programming for distributed data processing.	**BT 3**
CO 3	Apply Hadoop architecture, HDFS, and MapReduce programming for distributed data processing.	**BT 4 & 5**
CO 4	Design and optimize NoSQL database systems (e.g., MongoDB, Cassandra) for big data applications.	**BT 6**

Detailed Syllabus:

Module	Topics	Course Content	Periods

I.	**Introduction to Big Data and Storage Systems**	Introduction to Big Data: Definition and Characteristics (3Vs: Volume, Velocity, Variety), Challenges in Big Data Analytics, Applications in Healthcare, Finance, and IoT, Big Data Storage & Management, Traditional Databases vs. Big Data Storage, NoSQL Databases (MongoDB, Cassandra, HBase), Distributed File Systems (HDFS, Amazon S3, Google Bigtable), Data Acquisition & Preprocessing, Data Ingestion: Batch vs. Stream Processing, Data Cleaning and Transformation, Schema Design for Big Data, Introduction to Distributed Computing, Basics of Parallel and Distributed Processing, CAP Theorem and BASE Properties, Google's Big Data Technologies: Bigtable, MapReduce, Spanner	**22**
II.	**Hadoop & Spark**	Hadoop Ecosystem, Hadoop Architecture and Components (HDFS, YARN, MapReduce), Hadoop Cluster Setup, Hadoop vs. Spark, MapReduce Programming Model, Understanding the MapReduce Workflow, Writing MapReduce Programs (Java/Python), Combiner and Partitioner in MapReduce, Apache Spark & Resilient Distributed Datasets (RDDs), Spark Core Concepts and Architecture Transformations and Actions in RDDs, Spark DataFrames and Datasets, Advanced Spark Concepts, Spark SQL and DataFrames, Spark MLib for Machine Learning, Performance Tuning in Spark	**22**
III.	**Machine Learning & Streaming Analytics**	Machine Learning with Big Data, Challenges of Machine Learning on Big Data, Scalable ML Algorithms (Decision Trees, Clustering, Regression), Apache Spark MLlib, Big Data Streaming Analytics, Introduction to Stream Processing, Apache Kafka, and Apache Flink Real-time Data Processing with Spark Streaming, Graph Processing with Big Data, Introduction to Graph Analytics, Apache Giraph and GraphX in Spark, PageRank Algorithm, Text & Social Media Analytics, Sentiment Analysis on Large-scale Text Data, Natural Language Processing (NLP) using Spark, Twitter and Social Media Data Analysis	**22**
IV	**Cloud-Based Big Data Analytics**	Big Data on Cloud Platforms, Google Cloud BigQuery, AWS Big Data Services (Redshift, EMR), Microsoft Azure Data Lake, Big Data Security & Privacy, Data Governance & Compliance (GDPR, CCPA), Secure Data Storage & Access Control, Ethical Considerations in Big Data Analytics, Big Data Use Cases & Applications, Fraud Detection in Banking & Finance, Healthcare Analytics for Disease Prediction Smart Cities and IoT Data Analysis, Future Trends in Big Data Analytics, AI and Big Data Integration, Quantum Computing for Big Data, Edge Computing and IoT Analytics	**22**
TOTAL			**88**

Big Data Analytics Practice Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Setup and configure Hadoop Distributed File System (HDFS).
- Perform CRUD operations on NoSQL databases (MongoDB, Cassandra).
- Implement batch and stream data ingestion techniques.
- Write a MapReduce program for word count and log processing.
- Implement data transformations using Spark RDDs and DataFrames.
- Perform SQL operations on Spark DataFrames
- Implement a recommendation system using Spark MLlib.
- Process real-time streaming data using Apache Kafka.
- Perform sentiment analysis on Twitter data.
- Deploy and analyze Big Data workloads on AWS/Azure.
- Perform fraud detection using Big Data techniques.
- Build a predictive model for healthcare analytics.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks

1. *Hadoop: The Definitive Guide,* Tom White, 3rd Edition, 2012, O'Reily
2. *Spark: The Definitive Guide,* Bill Chambers, Matei Zaharia, 1st Edition, 2017, O'Reily
3. *Mining of Massive Datasets,* Jure Leskovec, Anand Rajaraman, 2nd Edition, 2016, Dreamtech Press

Reference Books:

1. Nathan Marz, *Big Data: Principles and Best Practices of Scalable Real-Time Data Systems,* 1st Edition, 2015, Manning Publications
2. Mohammad Guller, *Big Data Analytics with Spark*, 1st Edition, 2015, Apress

6.4 Detailed Syllabus for 3rd Semester:

Paper I/ Subject Name: Network Security and Cryptography		Subject Code: CAP054C301
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to make the students understand fundamental Network security and cryptography concepts

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain the** fundamental concepts, applications, and ethical implications of Network Security & Cryptography	**BT 2**
CO 2	**Apply** Cryptography algorithms to solve problems	**BT 3**
CO 3	**Analyze** and implement network security protocols.	**BT 4**
CO 4	**Assess** and **design** Network Security & Cryptography Solutions	**BT 5 & 6**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Need for Security, Security Approaches, Principles of Security, Types of Attacks, Brute Force Attack, Encryption, Decryption, Cryptosystem, Cryptographic Techniques: Substitution Ciphers, Transposition Ciphers, Product Ciphers, Steganography, Block Cipher, Stream Cipher.	**22**
II	**Symmetric and Asymmetric Key Cryptography**	Overview, Algorithm Modes and Types, Data Encryption Standard: Simplified DES, The Strength of DES, Differential and Linear Cryptanalysis. Triple DES, Blowfish. Confidentiality using Conventional Encryption: Placement of Encryption Function, Traffic Confidentiality, Key Distribution, Random Number Generation. Modular Arithmetic, Public Key Cryptography and RSA: Principles of Public Key Cryptosystems, Difference with Symmetric Key Cryptography, The RSA Algorithms, Key Management, Diffie-Hellman Key Exchange.	**22**
III	**Authentication Protocols**	Message Authentication: Authentication Requirements, Authentication Functions, Message Authentication Codes, MD5 Message Digest Algorithms, Digital Signatures and Authentication Protocols: Digital Signatures, Authentication Protocols, Digital Signature Standards.	**22**

IV	**Security Protocols**	Security Applications and Protocols- Authentication Applications: Secure HTTP, HTTPS, ERT, SSH, Kerberos. Email Security: PGP, S/MIME. IP Security: Overview, IPSec architecture.	**22**
	TOTAL		**88**

Subject Name: Network Security and Cryptography Lab		**Subject Code: CAP054C312**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Total Practice Hours for the semester = 30 (2 hours per week)

Laboratory experiments based on the following-

1. Implementation of Caesar Cipher and Substitution Cipher
2. Encryption and Decryption using DES and AES
3. Simulating RSA Algorithm
4. Implement Diffie-Hellman Key Exchange
5. Hashing using SHA-256 and MD5
6. Digital Signature Implementation
7. Packet Sniffing and Analysis using Wireshark
8. Simulation of Firewall and VPN
9. SSL/TLS Handshake Demonstration
10. Setup and Configuration of IDS (e.g., Snort)

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks:

1. *Cryptography and Network Security*, Atul Kahate, 2nd Edition. 2003, Tata McGraw Hill.
2. *Cryptography and Network security,* Fourozan, 3rd Edition, 2007, McGraw Hill

Reference Books:

1. William Stallings, *Cryptography and Network Security*: *Principles and Practices*", 5th Edition, 2010, Prentice Hall.
2. Michael Howard, David LeBlanc, John Viega, *24 Deadly Sins of Software Security: Programming Flaws and How to Fix Them*, 1st Edition, 2009, McGraw-Hill Osborne Media.

Paper II/ Subject Name: Remote Sensing and GIS		**Subject Code: CAP054C206**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamental concepts of remote sensing and its applications, learn about GIS (Geographic Information Systems) and spatial data processing, explore satellite image acquisition, preprocessing, and classification techniques etc.

Prerequisites: Basics of Digital Image Processing, Linear Algebra & Probability, Python Programming

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the fundamentals of remote sensing and GIS.	**BT 2**
CO 2	**Process** and **interpret** satellite images for spatial analysis.	**BT 3**
CO 3	**Analyze** and **assess** GIS solutions for urban planning and disaster management.	**BT 4 & 5**
CO 4	**Design** AI/ML techniques for remote sensing image classification.	**BT 6**

Detailed Syllabus:

Module	**Topics**	**Course Content**	**Periods**
I.	**Remote Sensing Fundamentals**	Fundamentals of Remote Sensing, Definition & Historical Development, Electromagnetic Spectrum & Remote Sensing Principles, Energy Interactions with Atmosphere & Earth's Surface, Remote Sensing Platforms & Sensors, Satellite & Aerial Remote Sensing Systems, Optical, Infrared, Microwave, and Hyperspectral Sensors, Types of Satellites: Landsat, Sentinel, MODIS, LIDAR, Resolution in Remote Sensing, Spatial, Spectral, Temporal & Radiometric Resolutions, Sensor Characteristics and Their Applications, Remote Sensing Data Acquisition, Passive vs. Active Remote Sensing, Satellite Data Sources and Accessibility	**22**
II.	**Image Processing and Interpretation**	Preprocessing of Satellite Images, Radiometric & Geometric Corrections, Image Enhancement Techniques, Image Rectification & Registration, Image Classification Techniques, Supervised & Unsupervised Classification, Machine Learning Approaches in Image Classification, Object-Based Image Analysis (OBIA), Vegetation Indices & Environmental Applications, NDVI (Normalized Difference Vegetation Index), Land Use/Land Cover (LULC) Mapping, Change Detection Techniques, Thermal & Radar Remote Sensing, Thermal Infrared Remote Sensing, Microwave & SAR (Synthetic Aperture Radar) Imaging	**22**
III.	**Geographic Information System (GIS)**	Fundamentals of GIS, GIS Concepts, Components & Data Models, Spatial Data Representation (Vector & Raster Data), GIS Software (ArcGIS, QGIS, Google Earth Engine), Spatial Data Acquisition & Integration, GPS (Global Positioning System) & Field Data Collection, Remote Sensing Data Integration with GIS, Spatial Analysis &	**22**

		Modeling, Buffering, Overlay, and Proximity Analysis, Network Analysis & Terrain Modeling, 3D GIS and DEM (Digital Elevation Model), Web GIS & Cloud-Based GIS Services, Google Earth Engine & OpenStreetMap, Cloud GIS Technologies (ArcGIS Online, Google Earth Engine)	
IV	**Applications**	Environmental & Agricultural Applications, Deforestation & Land Degradation Monitoring, Crop Yield Estimation & Precision Agriculture, Urban & Disaster Management, Urban Growth Analysis & Smart Cities, Flood, Earthquake, and Forest Fire Mapping, Climate Change & Hydrological Applications, Glacier & Coastal Change Detection, Watershed Management & Hydrological Modeling, Artificial Intelligence & Deep Learning in Remote Sensing, AI-Based Image Segmentation, Deep Learning for Land Cover Classification, Real-Time Remote Sensing Applications	**22**
TOTAL			**88**

Remote Sensing and GIS Practice Sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Download and analyze Landsat/Sentinel satellite images.
- Explore spectral bands and their applications.
- Visualize remote sensing data using GIS software (QGIS/ArcGIS).
- Perform radiometric and geometric corrections on satellite imagery.
- Implement NDVI for vegetation analysis.
- Classify land use using supervised and unsupervised learning methods.
- Create and analyze spatial data using QGIS/ArcGIS.
- Perform spatial interpolation and terrain modeling.
- Develop a simple Web GIS application.
- Perform flood risk analysis using GIS.
- Use machine learning models for land cover classification.
- Develop a GIS-based disaster monitoring system.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. *Remote Sensing and Image Interpretation,* Thomas M. Lillesand, Ralph W. Kiefer, Jonathan Chipman, 6th Edition, 2011, Wiley
2. *Introduction to Geographic Information Systems,* Kang-Tsung Chang, 4th Edition, 2017, McGraw Hill Education
3. *Fundamentals of Remote Sensing,* George Joseph, 3rd Edition, 2018, The Orient Blackswan

Reference Books:

1. John A. Richards, *Remote Sensing Digital Image Analysis*, 4th Edition, 2005, Springer

2. Peter A. Burrough, Rachael McDonnell, *Principles of Geographic Information Systems*, 3rd Edition, 2016, Oxford University Press

Paper III/Subject Name: Introduction to Deep Learning		**Subject Code: CAP054C303**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamentals of machine learning, apply supervised and unsupervised learning techniques, develop advanced machine learning models, explore deep learning architectures and algorithms, and design and train AI models using modern deep learning techniques.

Prerequisites: Linear Algebra, Probability & Statistics, Python Programming

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the key concepts of Deep Learning and their applications.	**BT 2**
CO 2	**Apply data-driven algorithms such as** regression, classification, and clustering.	**BT 3**
CO 3	**Analyze** and **assess** different neural network architectures and training techniques.	**BT 4 & 5**
CO 4	**Design** and implement deep learning models for real-world applications	**BT 6**

Detailed Syllabus:

Module	Topics	Course Content	Alignment with SDGs	Periods
I.	**ML Fundamentals**	Definition and Types of ML: Supervised, Unsupervised, Reinforcement Learning, Applications of ML in Healthcare, Finance, NLP, and Computer Vision, Overview of ML Pipelines, Linear Algebra: Vectors, Matrices, Eigenvalues, and Eigenvectors, Probability Theory: Bayes' Theorem, Conditional Probability, Optimization: Gradient Descent, Stochastic Gradient Descent (SGD), Linear Regression: Least Squares Method, Gradient Descent, Polynomial Regression, Ridge & Lasso Regression, Evaluation Metrics: MSE, RMSE, R^2 Score, Logistic Regression, k-Nearest Neighbors (k-NN), Decision Trees & Random Forest, Evaluation Metrics: Confusion Matrix, Precision, Recall, F1-Score	**SDG 4:** Quality Education (AI and ML foundations); **SDG 9:** Industry, Innovation and Infrastructure (data-driven intelligent systems); **SDG 3:** Good Health and Well-being (healthcare ML applications).	**22**
II.	**Advanced AL Techniques**	Support Vector Machines (SVM): Hard Margin & Soft Margin SVM Kernel Trick: RBF, Polynomial Kernels, Unsupervised Learning, Clustering: k-Means, Hierarchical Clustering, DBSCAN, Dimensionality Reduction: Principal Component Analysis (PCA), t-SNE, Ensemble Learning & Boosting	**SDG 8**: Decent Work and Economic Growth (advanced AI skills for industry); **SDG 9**: Industry, Innovation and	**22**

		Techniques, Bagging & Random Forest Boosting: AdaBoost, Gradient Boosting, XGBoost Neural Networks Basics, Perceptron & Multi-Layer Perceptron (MLP), Activation Functions: Sigmoid, ReLU, Tanh, Backpropagation Algorithm	Infrastructure (machine learning and intelligent automation).	
III.	**Deep Learning Fundamentals**	Introduction to Deep Learning, Difference Between ML and DL, Applications of Deep Learning (NLP, Image Recognition, Generative Models), Neural Networks & Optimization, Deep Neural Networks (DNN): Weight Initialization, Vanishing & Exploding Gradient Problems, Optimizers: SGD, Adam, RMSprop, Convolutional Neural Networks (CNNs), Convolution & Pooling Layers: Popular CNN Architectures: LeNet, AlexNet, VGG, ResNet, Recurrent Neural Networks (RNNs) & Sequence Models, RNNs & Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Applications in NLP & Time-Series Forecasting	**SDG 3**: Good Health and Well-being (medical image analysis and diagnostics); **SDG 9**: Industry, Innovation and Infrastructure (deep learning technologies); **SDG 4:** Quality Education (advanced AI competencies).	**22**
IV	**Advanced DL Concepts**	Generative Models: Autoencoders & Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Transformers & Attention Mechanisms, Self-Attention and Multi-Head Attention, Transformer Architecture (BERT, GPT, T5), Reinforcement Learning Basics, Markov Decision Process (MDP), Q-Learning & Deep Q Networks (DQN), Ethics & Deployment of AI Models. Bias in AI Models, Fairness & Explainability, Model Deployment: Flask, FastAPI, TensorFlow Serving	**SDG 3:** Good Health and Well-being (medical image analysis and diagnostics); **SDG 9**: Industry, Innovation and Infrastructure (deep learning technologies); **SDG 4:** Quality Education (advanced AI competencies).	**22**
TOTAL				**88**

Introduction to Deep Learning Practice sessions

Total Practice Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Implement Linear and Polynomial Regression on a dataset.
- Implement Logistic Regression for a classification task.
- Apply k-NN and Decision Trees for classification and compare their performance.
- Implement SVM with different kernels.
- Perform k-Means clustering and PCA on real-world datasets.
- Apply Random Forest and boosting techniques for a classification problem.
- Implement a simple Deep Neural Network using TensorFlow/PyTorch.
- Train a CNN for image classification (MNIST/CIFAR-10).
- Build an RNN/LSTM model for sentiment analysis or stock price prediction.
- Implement a GAN for image generation.
- Fine-tune a pre-trained Transformer model for text classification.
- Deploy a deep learning model as an API using Flask or FastAPI.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. *Deep Learning,* Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2016, MIT Press
2. *Pattern Recognition and Machine Learning,* Christopher M. Bishop, 1st Edition, 2006, Springer

Reference Books:

1. Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, 2012, MIT Press
2. ***N**eural Networks and Deep Learning, Michael Nielsen, 2015, Determination Press*
3. Richard S. Sutton, Andrew G. Barto, *Reinforcement Learning: An Introduction,* 2nd Edition, 1998, Bradford Books
4. Michael Nielsen, *Neural Networks and Deep Learning*, 2010

Subject Name: Social Network Analysis		**Subject Code: CAP054C304**
L-T-P-C – 3-1-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objective of this course is to provide students with a comprehensive understanding of social networks, graph theory, network analysis techniques, and mining methods to analyze complex social interactions and information diffusion using computational approaches

Prerequisites: Data Structures and Algorithms, Database Management Systems, Basic Probability and Statistics

Course Outcomes:

SI No	Course Outcome	**BT Level**
CO 1	**Explain** the concepts, structure, and mathematical foundations of social networks and graph models.	**BT 1 & 2**
CO 2	**Apply** graph-based algorithms and network measures to analyze social network structures and interactions.sign and interface embedded hardware components.	**BT 3 & 4**
CO 3	**Analyze** communities, information diffusion, influence propagation, and link prediction in large-scale social networks.	**BT 5**
CO 4	**Evaluate** social network mining techniques, visualization methods, and ethical issues in real-world applications.	**BT 5**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I	**Network and Society**	Introduction to SNA: Definition, actors, ties, Need for SNA, Motivation and scope, Applications: Healthcare, social media, web and cyberspace, police and military, scientific research, Preliminaries: Graph basics, Types of Networks: Directed, weighted, signed, Link-centric View, Uni/Bipartite, signed, Node-centric View: Homogeneous, heterogeneous, Temporal & Local View: Dynamic, ego networks, Historical Development, Evolution of SNA, Visualization Tools: Gephi, NetworkX	**22**
II	**Network Measures**	Network Basics Degree, paths, Clustering Coefficient: Local/global, Connected Components: Connectivity, Degree Centrality Importance: Closeness Centrality, Shortest paths, Betweenness, Node/edge, Eigenvector, Influence: Katz & PageRank: Ranking, Hubs & Authorities: HITS, Assortativity, Mixing patterns, Transitivity & Reciprocity: Triads, Similarity: Structural/regular, Degeneracy: k-core, coreness	**22**
III	**Network Growth Models and Link Analysis**	Real-world Properties: Small world, Random Networks: ER model, Degree Distribution: Poisson, Ring Lattice Model, Watts-Strogatz, Formation, Preferential Attachment: BA model, Price Model Citation, Local-world Model, Growth, Accelerating Growth, Evolution, Aging Models, Time effects. Link Analysis: Applications, Overview, Signed Networks, Balance, status, Strong/Weak Ties, Triadic closure, Link Algorithms Overview, PageRank, Random walk, Personalized PageRank, Variants, DivRank: Vertex reinforced, SimRank: Similarity, PathSim Meta-path, Problem solving	**22**
IV	**Community Structures in Networks and Link Prediction**	Community Structures in Networks: Applications, Overview, Types, Communities, Detection Methods: General, Disjoint Detection Node/modularity, Overlapping, Clique, Link partition, BigClam Model, Local Community, Local modularity, Evaluation, Ground truth, Case studies Link Prediction: Applications, Overview, Temporal Changes, Dynamics, Problem Definition, Evaluation, Heuristic Models Local/global, Probabilistic Models, Hierarchical, Supervised Random Walk, Framework, Information Theory, Mutual information, Latest Trends, Embeddings, GNN	**22**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	-	4 * 8 NCH = 32 NCH

Textbook

1. *Social Network Analysis*, Tanmoy Chakraborty, 2021, Willey
2. *Networks, Crowds, and Markets,* Easley & Kleinberg, 2010, Cambridge University Press.

Reference Books

1. Newman, *Networks: An Introduction*, 1st Edition, 2010, Oxford University Press.
2. Charu C. Aggarwal, *Social Network Data Analytics*, 2011, Springer.

6.5 Detailed Syllabus for 4th Semester

Paper I/ Subject Name: Quantum Computing		**Subject Code: CAP054C401**
L-T-P-C – 3-1-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

To introduce undergraduate students to the fundamental principles of quantum computing, quantum mechanics for computation, and the development of quantum algorithms with exposure to quantum programming frameworks.

Prerequisites: Linear Algebra and Basic Probability, Discrete Mathematics and Basic Programming Knowledge

Course Outcomes:

SI No	Course Outcome	BT Level
CO 1	Understand fundamental principles of quantum mechanics relevant to computing	**BT 1 & 2**
CO 2	Apply quantum gates and circuits to simple computational problems	**BT 3 & 4**
CO 3	Analyze and simulate quantum algorithms	**BT 4 & 5**
CO 4	Develop basic quantum programs using frameworks like Qiskit	**BT 5**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I	**Quantum Computation Foundations**	Introduction to quantum computing. Differences from classical computing. Quantum bits (qubits), quantum states, Dirac notation, superposition and entanglement.	**22**
II	**Quantum Gates and Circuits**	Single and multi-qubit gates (Pauli, Hadamard, Phase, CNOT, Toffoli). Quantum circuit representation and simplification. Quantum measurement and state collapse.	**22**
III	**Quantum Algorithms**	Deutsch-Jozsa algorithm, Grover's search algorithm, Shor's factoring algorithm. Quantum teleportation and superdense coding. Overview of quantum supremacy and limitations.	**22**
IV	**Quantum Programming and Applications**	Introduction to Qiskit and IBM Quantum Experience. Creating and simulating quantum circuits. Applications in cryptography, optimization, and machine learning.	**22**

Quantum Computing Practice Session

Total Practice Hours for the semester = 30 (2 hours per week)

Laboratory experiments based on the following-

1. 1. Simulate single and multiple qubit operations using Qiskit.

2. Construct and analyze quantum gates and circuits.

3. Implement Deutsch-Jozsa algorithm using IBM Q platform.

4. Run Grover's search on a simple dataset.

5. Perform quantum teleportation using Qiskit simulator.

6. Simulate quantum entanglement and measurement outcomes.

7. Visualize Bloch sphere representations of qubit states.

8. Create a quantum coin toss experiment.

9. Explore noise and decoherence using Qiskit Aer.

10. Mini project: Quantum algorithm simulation for real-world problem.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH(Optional)	

Textbook

1. Quantum Computation and Quantum Information, Michael A. Nielsen and Isaac L. Chuang, 10th Anniversary Edition, 2010, Cambridge University Press.

Reference Books:

1. Quantum Computing: An Applied Approach, Jack D. Hidary, 1st Edition, 2019, Springer.

2. Learn Quantum Computing with Python and Q#, Sarah Kaiser and Christopher Granade, 1st Edition, 2021, Manning Publications.

3. Programming Quantum Computers, Eric R. Johnston et al., 1st Edition, 2019, O'Reilly.

Paper II/ Subject Name: Blockchain Technologies		**Subject Code: CAP054C402**
L-T-P-C – 3-1-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

To introduce students to the principles, architecture, and applications of blockchain technologies. The course will provide knowledge on decentralized systems, consensus algorithms, cryptocurrency, and smart contracts with hands-on exposure to blockchain development environments

Prerequisites: Basics of Computer Networks and Cryptography, Programming fundamentals

Course Outcomes:

SI No	Course Outcome	BT Level
CO 1	Understand blockchain architecture and distributed ledger technology	**BT 1 & 2**
CO 2	Analyze consensus mechanisms and smart contract frameworks	**BT 3 & 4**
CO 3	Apply blockchain principles in real-world applications	**BT 4 & 5**
CO 4	Evaluate the security, scalability, and governance aspects of blockchain	**BT 5**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I	**Introduction to Blockchain**	History and fundamentals of blockchain, structure of a block, characteristics of blockchain, public vs private vs consortium Blockchain, use cases across industries.	**22**
II	**Consensus Mechanisms and Cryptography**	Consensus algorithms: Proof of Work (PoW), Proof of Stake (PoS), Practical Byzantine Fault Tolerance (PBFT), Cryptographic principles: hash functions, digital signatures, Merkle trees.	**22**
III	**Smart Contracts and Ethereum**	Ethereum architecture, EVM, Solidity programming basics, developing and deploying smart contracts, ERC-20 tokens, Gas, Remix IDE.	**22**
IV	**Blockchain Applications and Challenges**	Blockchain in supply chain, finance, healthcare, identity management. Scalability and interoperability issues. Security, privacy, regulatory and governance challenges.	**22**

Blockchain Technologies Practice Session

Total Practice Hours for the semester = 30 (2 hours per week)

Laboratory experiments based on the following-

1. Set up and explore a local Ethereum test network using Ganache.
2. Create and deploy a simple smart contract using Solidity.
3. Implement a basic cryptocurrency token using ERC-20 standard.
4. Interact with smart contracts using Web3.js or ethers.js.
5. Build a decentralized voting application on Ethereum.
6. Configure a private blockchain network using Hyperledger Fabric.
7. Evaluate blockchain performance using block explorers and metrics.

8. Simulate double-spending and analyze blockchain defense mechanisms.

9. Use Metamask for interacting with Ethereum DApps.

10. Mini Project: Develop a full-stack decentralized application (DApp).

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH(Optional)	

Textbook:

1. Mastering Blockchain, Imran Bashir, 3rd Edition, 2020, Packt Publishing.

Reference Books:

1. Blockchain Basics: A Non-Technical Introduction, Daniel Drescher, 1st Edition, 2017, Apress.

2. Blockchain Applications: A Hands-On Approach, Arshdeep Bahga and Vijay Madisetti, 1st Edition, 2017, VPT.

3. Ethereum: Blockchains, Digital Assets, Smart Contracts, Andreas M. Antonopoulos and Gavin Wood, 1st Edition, 2022, O'Reilly.

Paper III/ Subject Name: Soft Computing **Subject Code: CAP054C403**

L-T-P-C – 3-1-0-4 **Credit Units: 04** **Scheme of Evaluation: T**

Objective:

To introduce students to the fundamentals of soft computing techniques, including fuzzy logic, genetic algorithms, neural networks, and hybrid systems. The course aims to develop the ability to design and implement intelligent systems capable of learning and adaptation.

Prerequisites: Basic Programming Knowledge, Fundamentals of Mathematics and Logic

Course Outcomes:

SI No	Course Outcome	BT Level
CO 1	Understand the concept of soft computing and its applications	**BT 1 & 2**
CO 2	Apply fuzzy logic principles for problem solving	**BT 3**
CO 3	Design neural networks and analyze their performance	**BT 4 & 5**
CO 4	Implement genetic algorithms for optimization	**BT 5 & 6**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I	**Introduction to Soft Computing & Fuzzy Logic**	Definition and characteristics of soft computing. Introduction to fuzzy sets and fuzzy logic, fuzzy membership functions, fuzzy rules and fuzzy inference, fuzzy decision-making.	**22**
II	**Artificial Neural Networks**	Introduction to neural networks, biological neuron, perceptron model, multi-layer feedforward networks, backpropagation algorithm, training of neural networks, applications.	**22**
III	**Genetic Algorithms**	Introduction to evolutionary computation, genetic algorithm steps, selection, crossover, mutation, fitness function, convergence, applications of GA in optimization problems.	**22**
IV	**Hybrid Systems & Applications**	Neuro-fuzzy systems, genetic-fuzzy systems, ANN-GA integration. Case studies and real-world applications of soft computing in engineering and decision-making.	**22**

Soft Computing Practice Session

Total Practice Hours for the semester = 30 (2 hours per week)

Laboratory experiments based on the following-

- Implementation of Fuzzy Membership Functions: Create various membership functions: triangular, trapezoidal, Gaussian.
- Fuzzy Logic Control for Temperature System: Design a fuzzy controller for temperature regulation using fuzzy rules.
- Fuzzy Inference System using Mamdani Method: Build a Mamdani-type FIS for a real-life problem (e.g., fan speed controller).
- Fuzzy Inference System using Sugeno Method: Design a Sugeno-type FIS for mapping inputs to output decisions.
- Fuzzification and Defuzzification: Implement fuzzification and defuzzification for input-output mapping.
- Basic Perceptron Network :Implement a single-layer perceptron for logical operations (AND, OR).
- Multi-Layer Perceptron (MLP) with Backpropagation :Train a neural network for handwritten digit recognition (MNIST or subset).
- Activation Functions Visualization: Plot and compare sigmoid, tanh, and ReLU functions.
- Neural Network for XOR Problem: Solve the XOR classification using a 2-layer MLP.
- Implement Hebbian Learning Rule: Create a simple pattern recognizer using Hebbian learning.
- Implement Genetic Algorithm for Function Optimization: Use GA to maximize or minimize benchmark mathematical functions.
- Travelling Salesman Problem using GA:Apply a genetic algorithm to solve TSP for a small set of cities.
- Neuro-Fuzzy Inference System (ANFIS) :Design and train an Adaptive Neuro-Fuzzy system for a classification task.

Textbook:

1. Soft Computing and Intelligent Systems Design, F. O. Karry and C. De Silva, 1st Edition, 2004, Pearson Education.

Reference Books:

1. Neural Networks, Fuzzy Logic and Genetic Algorithms, S. Rajasekaran and G.A. Vijayalakshmi Pai, 1st Edition, 2003, PHI.

2. Principles of Soft Computing, S. N. Sivanandam and S. N. Deepa, 2nd Edition, 2011, Wiley India.

3. Introduction to Artificial Neural Systems, J. M. Zurada, 1st Edition, 1992, West Publishing Company.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH(Optional)	

Paper IV/ Subject Name: Edge Computing **Subject Code: CAP054C404**

L-T-P-C – 3-1-0-4 **Credit Units: 04** **Scheme of Evaluation: T**

Objective:

This course aims to provide students with a deep understanding of edge and fog computing paradigms, their architecture, communication models, and applications in real-time and latency-sensitive systems. Students will learn to design, deploy, and evaluate edge-based systems using platforms like Raspberry Pi, NVIDIA Jetson, or cloud-edge frameworks.

Prerequisites:

Prerequisites: Knowledge of computer networks, cloud computing, and basic IOT knowledge

Course Outcomes:

SI No	Course Outcome	BT Level
CO 1	Understand the fundamentals and architecture of edge and fog computing.	**BT 1 & 2**
CO 2	Analyze the role of edge computing in reducing latency and network congestion.	**BT 3 & 4**
CO 3	Design edge-enabled applications using lightweight computation models.	**BT 4 & 5**
CO 4	Evaluate different edge deployment architectures and data flow mechanisms.	**BT 5**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I	**Introduction to Edge Computing**	Motivation, evolution from cloud to edge; comparison of cloud, fog, and edge; edge devices, characteristics, use cases in smart cities, healthcare, and manufacturing.	**22**
II	**Architecture and Communication**	Edge architecture models: hierarchical vs. distributed, cloud-fog-edge continuum; edge gateways; communication protocols (MQTT, CoAP, HTTP, AMQP); latency and bandwidth considerations.	**22**
III	**Resource Management and Security**	Resource discovery, task scheduling, containerization at edge (Docker), load balancing; security & privacy challenges in edge, trusted execution environments (TEE), lightweight encryption, authentication models.	**22**
IV	**Edge Platforms and Applications**	Edge orchestration (KubeEdge, Azure IoT Edge), Open Horizon, TensorFlow Lite, edge ML models; case studies: autonomous vehicles, industrial IoT (IIoT), smart surveillance, edge analytics, streaming data processing.	**22**

Edge Computing Practice Session

Total Practice Hours for the semester = 30 (2 hours per week)

Laboratory experiments based on the following-

1. Set up an edge computing environment using a Raspberry Pi or emulator.
2. Implement MQTT-based data transmission from an IoT device to an edge server.
3. Build a latency comparison between edge and cloud processing using Python/Node.js.
4. Deploy a containerized application using Docker on an edge node.
5. Implement a face detection model using TensorFlow Lite at the edge.
6. Collect sensor data and process using edge analytics before cloud upload.
7. Configure KubeEdge for microservice orchestration.
8. Simulate fog computing using Cisco Packet Tracer or iFogSim.
9. Perform a comparative study on different communication protocols for edge.
10. Implement a security mechanism like token-based authentication in an edge application

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	2 * 15 NCH = 30 NCH(Optional)	

Textbook

1. Fog and Edge Computing: Principles and Paradigms, Rajkumar Buyya, Satish Narayana Srirama, 1st Edition, 2019, Wiley.

Reference Books

1. Edge Computing: From Hype to Reality, Perry Lea, 1st Edition, 2020, Packt Publishing.

2. Architecting the Cloud to Edge, Victor Muntés, 2023, Springer.

3. Edge AI, Xiaofei Wang and Min Chen, 2021, Springer.